Task 1: Understanding the Industry Problem

Project: Predictive Anemia Screening and Healthcare Resource Planning Using Full Blood Count Data in Sri Lankan Healthcare Institutions

Theme: Public Health Indicator Analysis

1.1 Industry Background

Healthcare is one of the largest data-generating industries in any economy, and Sri Lanka is no exception. The public health system provides free care at the point of delivery through a tiered network of institutions - from primary medical care institutions (PMCIs) up to teaching, provincial general, and district general hospitals - supported by a national laboratory network under the Ministry of Health.

Anemia, a condition defined by abnormally low hemoglobin (HB) concentration in the blood, remains one of the most persistent public health indicators tracked globally and in South Asia specifically. Earlier national data reported very high anemia prevalence in vulnerable groups - around 65% among pregnant women and 52.6% among children aged 5–10 in historical surveys [1]. More recent WHO Global Health Observatory data situates Sri Lanka more favorably within the South Asian region: while the region overall carries a burden exceeding 40% among women of reproductive age, Sri Lanka's comparatively lower rate has been attributed to stronger maternal nutrition and health system coverage [2]. World Bank figures placed prevalence among Sri Lankan women of reproductive age at roughly a third of that population as recently as 2019 [3]. Even recent community-based surveys of young adults find meaningful anemia burden once causes beyond classic iron deficiency are accounted for - iron deficiency was the single leading identified cause but accounted for a minority of cases, with additional burden from folate/B12 deficiency and haemoglobinopathies [4].

The Full Blood Count (FBC) - comprising hemoglobin, red cell indices (MCV, MCH, MCHC), packed cell volume, and red cell count - is the single most routinely ordered diagnostic test in both primary and secondary care. It is inexpensive, fast, and already collected at scale, making it a natural foundation for population-level health indicator analysis, unlike specialized tests (ferritin, B12, genetic

haemoglobinopathy panels) that are far less consistently available across the system.

1.2 Organizational Context

This project is framed as work commissioned by a Sri Lankan public healthcare organization (presentable in the viva as a Ministry of Health / Regional Director of Health Services-type client) operating across multiple institutions - mirroring the structure of our dataset, which spans four anonymized healthcare institutions.

Sri Lanka's public curative sector runs on a referral hierarchy: hospitals are categorized by resource allocation and availability, and patients are directed from primary to tertiary care institutions through this referral system [5]. Laboratory diagnostic capacity, however, is uneven across this hierarchy. A nationwide checklist-based assessment of primary medical care institutions found laboratory technicians were absent in roughly two-thirds of facilities assessed, and fewer than half of essential laboratory investigations were consistently available - largely due to reagent and consumable shortages and weak sample transport systems [6]. This resource unevenness is precisely the organizational backdrop that motivates a predictive, data-driven screening approach: where laboratory staff time and specialized testing are scarce, a model that flags likely anemia risk from already-collected FBC data offers a way to prioritize limited clinical and laboratory resources.

1.3 Description of the Business Problem

The organization currently relies on routine, manual interpretation of individual FBC results against WHO hemoglobin thresholds, on a patient-by-patient, encounter-by-encounter basis. There is no systematic, data-driven mechanism to:

- Identify which patient subgroups (by age, sex, institution) carry disproportionately high anemia risk;
- Flag patients are likely to be anemic to support earlier triage and prioritization;

- Aggregate case-level results into population-level indicators that could inform resource allocation (e.g., where to prioritize iron supplementation programs, staffing, or outreach).

In short: **large volumes of routinely collected FBC data are not being converted into actionable, forward-looking public health intelligence.** The organization is "data-rich but insight-poor" - a problem common across health systems that generate high case volumes but lack the analytical infrastructure to translate that data into strategic decisions.

1.4 Importance of Solving the Problem

- **Clinical significance:** Anemia is linked to fatigue, impaired cognitive development in children, adverse pregnancy outcomes, and reduced workforce productivity - it is not a benign incidental finding.
- **Resource significance:** Given documented shortages of laboratory technicians and essential investigations at the primary care level [6], any tool that helps target scarce diagnostic and treatment resources toward the highest-risk groups has direct operational value.
- **Equity significance:** Anemia burden is not evenly distributed by age, sex, or region; a systematic, evidence-based understanding of *where* risk concentrates support more equitable resource distribution rather than uniform, one-size-fits-all policy.
- **Strategic significance:** Sri Lanka's health system faces broader structural pressure - workforce migration of doctors and specialists has created staffing gaps, and shortages of essential drugs and laboratory facilities push patients toward costlier private-sector care despite the policy of free healthcare [7]. Under these constraints, low-cost, already-available data like FBC results becomes strategically more valuable, since it requires no new investment to collect - only better analysis.

1.5 Stakeholders Affected

Stakeholder	Interest / Impact
Patients	Timelier identification of anemia risk; earlier referral and treatment
Clinicians & lab technicians	Decision support that helps prioritize workload under staffing shortages
Hospital / institution management	Evidence base for internal resource and staffing decisions
Ministry of Health / regional health authorities	Population-level indicators to guide public health programs (e.g., nutrition, supplementation campaigns) and cross-institution resource allocation
Public health researchers	A reusable, evidence-based model of anemia risk factors specific to the Sri Lankan context
Policy makers / donors (e.g., World Bank-funded health system strengthening projects)	Justification and monitoring data for continued investment in laboratory and primary care capacity

1.6 Business Decisions Influenced by the Problem

- **Clinical triage:** which patients should be prioritized for confirmatory testing or treatment when staff time is limited.
- **Resource allocation:** where to direct laboratory staffing, reagent stock, and supplementation programs across institutions and regions.

- **Screening policy:** whether routine, low-cost FBC-based risk flags could reduce reliance on additional specialized (and scarcer) diagnostic tests.
- **Monitoring & evaluation:** how to track anemia burden over time and across institutions as a public health indicator, to evaluate whether interventions are working.

1.7 Consultancy Objectives

Our consultancy engagement has four objectives, mapped directly onto the remaining tasks of this project:

1. **Understand the data-generating problem** - assess data quality, distribution, and initial patterns in the FBC dataset (Task 3).
2. **Establish statistical evidence** for how anemia prevalence and severity vary across sex, age group, and institution (Task 4).
3. **Develop and critically evaluate a predictive screening model** for anemia risk using routinely collected FBC and demographic variables - deliberately excluding institution identity as a predictor to preserve generalizability beyond the four institutions studied (Task 5).
4. **Translate statistical findings into a concrete innovation proposal** for healthcare resource planning — e.g., a risk-flagging / early-warning framework usable at the point of care or for regional resource allocation (Task 10) - validated against real-world clinical/public-health expert feedback (Task 11).

Sources cited in this section

1. Health status, trends, and issues in Sri Lanka. ResearchGate. https://www.researchgate.net/publication/6152070_Health_status_trends_and_issues_in_Sri_Lanka

2. Global and South Asian trends in the prevalence of anemia among women of reproductive age (2010–2023): a review of WHO data with evidence from Pakistan. *Journal of Public Health*, Springer Nature. https://link.springer.com/article/10.1007/s10389-026-02775-3

3. Sri Lanka – Prevalence of anemia. IndexMundi (World Bank data). https://www.indexmundi.com/facts/sri-lanka/prevalence-of-anemia

4. Beyond iron deficiency: A comprehensive national survey of anaemia etiology in Sri Lankan young adults. *Scientific Reports* / PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC13136422/

5. HAPI-FHIR Server Implementation to Enhancing Interoperability among Primary Care Health Information Systems in Sri Lanka. arXiv. https://arxiv.org/pdf/2402.02838

6. Primary Health Care System Strengthening Project in Sri Lanka: Status and Challenges with Human Resources, Information Systems, Drugs and Laboratory Services. PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC9691080/

7. Reforming Healthcare in Sri Lanka for Equity and Excellence. Jaffna Monitor. https://www.jaffnamonitor.com/reforming-healthcare-in-sri-lanka-for-equity-and-excellence/